

Sukhrob Ilyosbekov

Computer Vision · Deep Learning · Explainable AI

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RESEARCH INTERESTS

I'm drawn to the gap between a vision model that scores well and one people can actually rely on. I first ran into it in medical imaging, where a skin-lesion classifier can be accurate and still be looking at the wrong part of the image, with no easy way for a clinician to notice. The same question has followed me into cell microscopy and generative image editing. In a PhD I'd like to work on representation learning and evaluation for medical and scientific images, where labels are expensive to get and a confident mistake matters more than a leaderboard number. Building models for patient care in industry is a big part of why I care about this.

EDUCATION

Northeastern University, Boston, MA

January 2024 – May 2026

- Master of Science in Computer Science; GPA 4.0/4.0
- Coursework: Advanced Perception, Computer Vision, Machine Learning

Suffolk University, Boston, MA

September 2021 – May 2023

- Bachelor of Science in Computer Science; Cum Laude; GPA 3.5/4.0
- Transferred from INTI International College, Malaysia (2019 – 2021) and Tashkent University of Information Technologies, Uzbekistan (2017 – 2019)

PUBLICATIONS

- [1] **S. Ilyosbekov**, S. Gajjar, and R. Jin, “MorphoCLIP: Text-Supervised Contrastive Learning for Perturbation Matching in Cell Painting Images,” *arXiv preprint*, arXiv:2608.22690, 2026. [\[Paper\]](#)
- [2] **S. Ilyosbekov**, “Localize, Don't Beautify: Client-Side Control of Image-Editing APIs for Cosmetic Surgery Previews,” *arXiv preprint*, arXiv:2608.02841, 2026. [\[Paper\]](#)
- [3] **S. Ilyosbekov**, “MelanomaNet: Explainable Deep Learning for Multi-Class Skin Lesion Classification,” *arXiv preprint*, arXiv:2512.09289, 2025. [\[Paper\]](#)

AWARDS & HONORS

- **Associate Member**, Sigma Xi, The Scientific Research Honor Society, 2026
- **Outstanding Project Award**, Bookshelf Scanner, Northeastern University
- **National Physics Olympiad, 1st Place**, Uzbekistan; first student from my high school to win nationally
- **University Coding Competition Winner**, Tashkent University of Information Technologies

RESEARCH

MorphoCLIP: Text-Supervised Cell Painting Retrieval

[GitHub](#) | [Paper](#)

PyTorch, DINOv3, BioClinical ModernBERT, contrastive learning, CPJUMP1

- Learns a shared space for Cell Painting images and text, so a compound or gene perturbation can be retrieved from a written description of it, and the other way round.
- DINOv3 and BioClinical ModernBERT stay frozen and only the projection heads train, which fits on a single consumer GPU.
- Evaluated on CPJUMP1: 51 plates, 3M+ images, 303 compounds, and 160 genes. Batch correction in embedding space keeps the model from simply recognizing plates.
- First author, with S. Gajjar and R. Jin. I led model design, training, and evaluation.

Localize, Don't Beautify: Client-Side Control of Image-Editing APIs

[Paper](#)

Image-editing APIs, inpainting models, ArcFace, prompt engineering

- Studies how far a client can control a black-box editing API, using cosmetic-surgery previews as the test case.

- I ran prompt steering, masked compositing, and model-based inpainting through six commercial editors on 196 facelift and rhinoplasty edits.
- Masked compositing, the simplest of the three, kept edits localized best, with identity preservation scored by ArcFace. Sole author.

MelanomaNet: Explainable Deep Learning for Skin Lesion Classification

[GitHub](#) | [Paper](#)

PyTorch, EfficientNet V2, focal loss, GradCAM++, OpenCV, ISIC 2019

- EfficientNet V2 (384×384, focal loss) over all nine ISIC 2019 classes, reaching 0.86 weighted F1 on 25K dermoscopic images.
- Splits GradCAM++ attention along the ABCDE criteria. Asymmetry, border irregularity, color variation (K-means), and diameter come straight from the lesion mask, and the paper scores how closely the model's attention agrees with each one. Sole author.

SELECTED PROJECTS

LightDepth: Lightweight Monocular Depth Estimation

[GitHub](#)

PyTorch, ResNet18, U-Net, NYU Depth V2

- ResNet18 encoder and U-Net decoder for single-image depth. It has 42% fewer parameters than Depth Anything V2 (14.3M vs. 24.8M) and runs 72% faster.
- Error on NYU Depth V2 is lower too: AbsRel down 5.3% and SqRel down 31.1% against the same baseline.

FSRCNN: Accelerating Super-Resolution Convolutional Neural Network

[GitHub](#)

PyTorch, mixed-precision training, T91, Set5, Set14, DIV2K

- Reimplemented FSRCNN (Dong et al., ECCV 2016) at 2×, 3×, and 4×, with upsampling learned end to end.
- Recovered the paper's improvement over SRCNN (+1.78 dB PSNR on Set5, +1.26 dB on Set14), then ran ablations on the shrinking and mapping layers.

Bookshelf Scanner: Multi-Modal Book Detection and Recognition

[GitHub](#)

YOLO, Moondream2 VLM, llama.cpp, PyTorch, FastAPI

- YOLO instance segmentation finds each spine in a shelf photo, and Moondream2 reads the title and author.

WORK EXPERIENCE

Machine Learning Engineer | [EmTech Care Labs](#) | Portland, ME

Jan 2025 – Present

- Built a model that uses daily activity and caregiver logs to flag Alzheimer's residents at risk of decline, so staff can step in earlier.
- Wrote an NLP pipeline that turns free-text clinical notes into structured records of symptoms, medications, and care events. Both models run in production on de-identified data under HIPAA.

Machine Learning Engineer | [AllFactors](#) | Santa Clara, CA

Oct 2021 – Dec 2024

- Built the property-valuation and market-forecast models that replaced the company's rules-based system.
- Wrote the real-time feature pipeline that fed them (SignalR, SQL Server change tracking) and the Blazor dashboards where users saw the results.

Earlier: Software Engineer, Smart Meal Service, Russia (2020 – 2021). Game Developer, Pentalight Technology, Malaysia (2020 – 2021). Freelance (2019 – 2020). Game Developer, EC Dev Team, Uzbekistan (2016 – 2019).

TECHNICAL SKILLS

Deep Learning & CV: PyTorch, TensorFlow/Keras, OpenCV, CNNs, Vision Transformers, DINOv3, U-Net, EfficientNet, YOLO, contrastive and self-supervised learning, GradCAM++, detection and segmentation, depth estimation

Mathematics: Linear algebra, probability and statistics, convex optimization, numerical methods, information theory, hypothesis testing and experimental design

Machine Learning: Transfer learning, loss design, regularization, hyperparameter search, cross-validation and error analysis, XGBoost, K-means, NLP

Research Computing: Python, NumPy/SciPy, pandas, scikit-learn, Hugging Face, CUDA, mixed-precision and multi-GPU training, Weights & Biases, Docker, $\LaTeX$

Software Engineering: C#, C++, TypeScript; .NET, FastAPI, PostgreSQL, Kubernetes, AWS, Azure